

MOMENTUM: $p = mv$

- Impulse (change of momentum): $\Delta p = F \Delta t$ (F is average force)

WORK: $W = f * d$ (times cosine θ)

- Power (change of work over time): $W/\Delta t$

Kinetic Energy: $\frac{1}{2}mv^2$

Gravitational Potential energy is negative???

Potential energy curves are **negative** integral of Force

Centripetal Acceleration: $v^2 / r = \omega^2 * r$

- remember, force is multiplied by m

SHM & torque:

- Period: $T = 2\pi \omega = 2\pi \sqrt{k/m}$
- $f = 1/T$
 - $x = A \cos(2\pi * f * t)$
 - $v = -A \omega \sin(2\pi * f * t)$
 - $a = -A \omega^2 \cos(2\pi * f * t)$
- PERIOD OF SIMPLE PENDULA: $2\pi * \sqrt{L/g}$
 - PERIOD OF PHYS PEND: $2\pi * \sqrt{I/mgL}$
- Angular Momentum: L (different L) = $\omega * m = I * \omega$ (if inertia)
- Moment of Inertia: $I = \text{sum of } MR^2 = \text{specifics in some cases}$
- Rotational Kinetic Energy: $\frac{1}{2} I \omega^2$
- KEPLERS: $T^2 = (4\pi^2) / (G * M_{\text{total}}) * a^3$
 - Gravitation Force: $F = (G * m_1 * m_2) / r^2$

Adjacent springs have constants combined

Sequential has $1/k_1 + 1/k_2 = 1/k_{\text{eff}}$

Parallel Axis Theorem: $I_{\text{new}} = I_{\text{center}} + Mh^2$

ROCKETRY: $\Delta v = u \ln(m_i / m_f)$ [u is speed of gas]

remember to revisit drag forces

